

Force Plate

Plug USB into laptop (or USB hub)
Plug other end (printer cable looking end) into force plate
Open NetForce on laptop
Startup > Hardware Zero
Close NetForce

fNIRS

Plug fNIRSport2 box into wall charger if needed
Connect all sources and detectors to the fNIRS
Press power button on fNIRSport2
Confirm: Power light = GREEN and F2 light = GREEN

EEG

Charge LiveAmp device if needed
Attach BrainVision USB dongle to laptop (or USB hub)
Turn on BrainVision LiveAmp (may need to hold down the button for a few seconds)

IMUs

Plug the Awinda Station into the laptop/USB hub
Place one (or two or however many) IMU sensors into Awinda docking station

Arduino Buttons

Plug button box into laptop/USB hub (USB connection)
Open Arduino IDE
Load correct Arduino script (should already be there when you open it up)
Verify and upload code to board

Hammer Orientation App

Open iPad
Click the app that says Open in Safari (picture of a hammer)
Place iPad in front of subject
On the laptop or a phone or literally any device, open the Hammer controller app
If you click the Next button, the hammer should change orientation on both controller and display devices

Laptop

Connect to WiFi: [nsp2_2422_0836_A](#)
Connect to LiveAmp via Bluetooth (just click it on the computer even though it may not register yet...its fine)

Open Visual Studio Code
You should see two files open at the top
["RecorderForm.cs"](#)
"OGpython_with_IMU.py"
Don't do anything yet

Launch Aurora 2023.9

Click occmotor 16x16 (or whichever setup)
Click Edit
Click LSL Stream Names
Data Out Stream Name = **Aurora**
Trigger In Stream Name = **trigger**
Click Save
Click Select Config
Click Start Optimization of Signal Levels

Open LiveAmp - shortcut
Click Scan for devices
Wait for device: **055602-0308**
Click Link

(Optional but recommended check)
Open BrainVision LSL Viewer
Click Connect
Select LiveAmp
Confirm data streams appear
You won't see markers appear as LSL only allows itself to be seen by one entity

Open LabRecorder - shortcut
Confirm LiveAmp stream is visible
Do NOT press Start recording yet

Go back to Aurora
Open Recording Page
Start Recording
Start Experiment

Go to VS Code
Run "[RecorderForm.cs](#)" by typing dotnet run in the terminal
A pop up will appear waiting for Python
Run the Python experiment script "OGpython_with_IMU.py" by clicking the play button in the top right corner
Run python but do not click start yet!

Go to LabRecorder
Click Update
Select: **bv_markers and Aurora**
Select: **all entries**
Click Start

Return to Python script window
Click Start
Follow any on-screen instructions

Have subject stay on plate whole time (dont have them get on and off)

Stop Python script
Stop Aurora
Stop BrainVision recording
In lab recorder, take note of where it saved to (a .xdf file)
Unlink LiveAmp in connector software
Close LiveAmp connector
Close Python GUI
Close Aurora
stop/close IMU window
See notes in python terminal

After running script, see results folders on desktop.....

python code saved as .xlsx

C:\Users\rpier12\Python Results\button_task_log_20260603_112736.xlsx

EEG data file saved as .xdf

C:\Users\rpier12\Documents\CurrentStudy\sub-P001\ses-S001\eeeg\sub-P001_ses-S001_task-Default_run-001_eeg.xdf

fNIRS data file saved as file folder containing multiple files

C:\Users\rpier12\Documents\NIRx\Data\2026-06-03

foot force plate data file saved as .txt

C:\AMTI\PythonResults

IMU data files saved as two .csv and one .mtb per session

C:\Users\rpier12\Documents\IMUData